

CHAPTER 5. INTEGRATION

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5.1 Antidifferentiation

Definition 1 (Antiderivative)

A function F is called an **antiderivative** of f on an interval I if

$$F'(x) = f(x) \quad \text{for all } x \in I.$$

Theorem 2

If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is $F(x) + C$, where C is an arbitrary constant.

NOTATION: The notation

$$\int f(x) dx = F(x) + C$$

is called the **indefinite integral of f** and C is called the **constant of integration**.

Example 3

Find the most general antiderivative of each of the following functions

$$(a) f(x) = \sin x \quad (b) f(x) = \frac{1}{x} \quad (c) f(x) = x^n, \quad n \neq -1$$

Function	Particular Antiderivative	Function	Particular Antiderivative
$cf(x)$	$cF(x)$	$\sin x$	$-\cos x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec^2 x$	$\tan x$
$x^n \quad (n \neq -1)$	$\frac{x^{n+1}}{n+1}$	$\csc^2 x$	$-\cot x$
$\frac{1}{x}$	$\ln x$	$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
e^x	e^x	$\frac{1}{1+x^2}$	$\tan^{-1} x$
$\cos x$	$\sin x$	$\frac{1}{ x \sqrt{x^2-1}}$	$\sec^{-1} x$

Example 4

Evaluate

a. $\int (x^3 - 6x) dx$

b. $\int (5\sqrt{x} + 4 \sin x) dx$

Applications of Antiderivative

Recover function from its derivative

If f is an antiderivative of f' , then

$$f(x) = \int f'(x) dx \quad (1)$$

Example 5

The graph of a certain function f has slope $4x^3 - 5$ at each point (x, y) and contains the point $(1, 2)$. Find the function f .

Applications of Antiderivative

Rectilinear Motion

If the object has position function $s = s(t)$, then

$$s'(t) = v(t) \implies s(t) = \int v(t) dt$$

$$v'(t) = a(t) \implies v(t) = \int a(t) dt$$

Example 6

A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6$ cm/s and its initial displacement is $s(0) = 9$ cm. Find its position function $s(t)$.

Example 7

The brakes of a certain automobile produce a constant deceleration of 22 ft/s^2 . If the car is traveling at 60 mi/h (88 ft/s) when the brakes are applied, how far will it travel before coming to a complete stop?

5.2 Area as the Limit of a Sum

Example 8

Calculate the area under the line $y = x$ and above the x -axis on the interval $[0, 1]$

- using area formula
- expressing area as the limit of a sum

SUMMATION FORMULAS

$$1. \sum_{k=1}^n k = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

$$2. \sum_{k=1}^n k^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$3. \sum_{k=1}^n k^3 = 1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$$

5.3 Riemann Sums and the Definite Integral

Problem 9 (The Distance Problem)

Suppose the odometer on our car is broken and we want to estimate the distance driven over a 30-second time interval. We take speedometer readings every five seconds and record them in the following table:

Time (s)	0	5	10	15	20	25	30
Velocity (mi/h)	17	21	24	29	32	31	28

Notice that $1 \text{ mi/h} = 5280/3600 \text{ ft/s}$.

Time (s)	0	5	10	15	20	25	30
Velocity (ft/s)	25	31	35	43	47	46	41

QUESTION

Assume that $v = f(t)$ for $a \leq t \leq b$, where f is positive and continuous on $[a, b]$. Find the distance traveled over time interval $[a, b]$?

ANSWER: The distance traveled over time interval $[a, b]$

$$s = \int_a^b f(t)dt \quad (2)$$

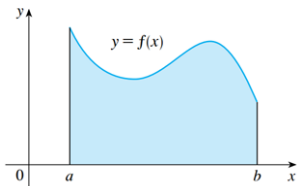
Some Notes about Integral

- ◇ Notation $\int_a^b f(x)dx$
 - $f(x)$ is called the **integrand**
 - a and b are called the **limits of integration**; a is the **lower limit** and b is the **upper limit**
 - The symbol dx simply indicates that the independent variable is x .
- ◇ The definite integral $\int_a^b f(x)dx$ is a number; it does not depend on x .

$$\int_a^b f(x)dx = \int_a^b f(t)dt = \int_a^b f(r)dr$$

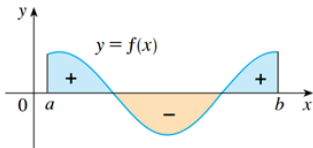
Some Notes about Integral

- ◇ If $f(x) \geq 0$ for $a \leq x \leq b$, then $\int_a^b f(x)dx$ is the area under the curve $y = f(x)$ above the x -axis from a to b .



- ◇ If f takes on both positive and negative values. Then a definite integral can be interpreted as a net area, that is, a difference of areas

$$\int_a^b f(x)dx = A_1 - A_2$$



Example 10

Evaluate the following integrals by interpreting each in terms of areas.

$$(a) \int_0^1 \sqrt{1-x^2} dx$$

$$(b) \int_0^3 (x-1) dx$$

Properties of the Integral

1. $\int_a^b c dx = c(b - a)$, where c is any constant
2. $\int_a^b cf(x) dx = c \int_a^b f(x) dx$, where c is any constant
3. $\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$
4. $\int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$
5. If $f(x) \geq 0$ for $a \leq x \leq b$, then $\int_a^b f(x) dx \geq 0$.
6. If $f(x) \geq g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$.
7. If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then

$$m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$$

5.4 The Fundamental Theorem of Calculus

The First Fundamental Theorem of Calculus

If f is continuous on $[a, b]$ and F is any antiderivative of f , then

$$\int_a^b f(x) dx = F(b) - F(a) \quad (3)$$

Example 11

Evaluate the integral

a. $\int_1^3 e^x dx$

b. $\int_3^6 \frac{1}{x} dx$

Example 12

- (a) Find the area under the parabola $y = x^2$ from 0 to 1.
- (b) Find the area under the cosine curve from 0 to $\frac{\pi}{2}$.

HISTORY: the French mathematician Gilles de Roberval first found the area under the sine and cosine curves in 1635, this was a very challenging problem until the discovery of the Fundamental Theorem by Barrow and then exploited by Newton and Leibniz in the 1660s and 1670s.

5.4 The Fundamental Theorem of Calculus

The Second Fundamental Theorem of Calculus

If f is continuous on $[a, b]$, then the function $F(x)$ defined by

$$F(x) = \int_a^x f(t) dt \quad a \leq x \leq b \quad (4)$$

is continuous on $[a, b]$ and differentiable on (a, b) , and

$$F'(x) = \frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x). \quad (5)$$

Example 13

Find the derivative of a function $g(x) = \int_0^x \sqrt{1+t^2} dt$.

Example 14

The **Fresnel function**

$$S(x) = \int_0^x \sin(\pi t^2/2) dt$$

is named after the French physicist **Augustin Fresnel** (1788–1827), who is famous for his works in optics.

Example 15

The **sine integral function**

$$Si(x) = \int_1^x \frac{\sin t}{t} dt$$

is important in electrical engineering.

Example 16

Find the derivative of a function $f(x) = \int_0^{x^2} t^2 dt$.

5.5 Integration by Substitution

Indefinite Integral

The Substitution Rule

Let $u = u(x)$ be a differentiable function of x and f be a continuous function. Then

$$\int f(u(x))u'(x)dx = \int f(u)du \quad (6)$$

Example 17

Find

a. $\int 2x\sqrt{x^2 + 1}dx$

b. $\int (2x - 1)^5 dx$

Example 18

Evaluate

a. $\int x^3 \cos(x^4 + 2) dx$

b. $\int (4 - 2 \cos t)^3 \sin t dt$

Example 19

Calculate

a. $\int e^{5x} dx$

b. $\int x e^{4-x^2} dx$

Example 20

Evaluate

a. $\int x(2x - 1)^3 dx$

b. $\int x^5 \sqrt{1 + x^2} dx$

Example 21

Show that

$$\int \tan x dx = -\ln |\cos x| + C \quad \text{and} \quad \int \cot x dx = \ln |\sin x| + C$$

Example 22

Water is flowing into a tank at the rate of $\sqrt{3t+1} \text{ ft}^3/\text{min}$. If the tank is empty when $t = 0$, how much water does it contain 5 min later?

Example 23

An oil storage tank ruptures at time $t = 0$ and oil leaks from the tank at a rate of $r(t) = 100e^{-0.01t}$ liters per minute. How much oil leaks out during the first hour?

5.5 Integration by Substitution

Definite Integral

The Substitution Rule for Definite Integral

If u' is continuous on $[a, b]$ and f is continuous on the range of $u = u(x)$, then

$$\int_a^b f(u(x))u'(x)dx = \int_{u(a)}^{u(b)} f(u)du \quad (7)$$

Example 24

Evaluate

a. $\int_0^4 \sqrt{2x+1} dx$

b. $\int_1^2 \frac{dx}{(3-5x)^2}$

c. $\int_1^e \frac{\ln x}{x} dx$

Integrals of Symmetric Functions

Suppose f is continuous on $[-a, a]$.

(a) If f is even, i.e., $f(-a) = f(a)$, then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx. \quad (8)$$

(b) If f is odd, i.e., $f(-a) = -f(a)$, then

$$\int_{-a}^a f(x) dx = 0. \quad (9)$$

Example 25

Evaluate

a. $\int_{-\pi}^{\pi} \sin x dx$

b. $\int_{-\pi/2}^{\pi/2} \cos x dx$

Example 26

Evaluate

a. $\int_{-1}^1 (x^6 + 1) dx$

b. $\int_{-1}^1 \frac{\sin x}{x^2 + 1} dx$

5.6 Introduction to Differential Equations

Introduction and Terminology

Definition 27 (Differential Equations, DEs)

An equation that contains a derivative or differential is called a **differential equation**.

$$F(x, y, y', y'', \dots, y^{(n)}) = 0. \quad (10)$$

The **order of a DE** is the highest order of derivative appearing in the equation.

- First order DEs:

$$\frac{dy}{dx} = 3x^2 + x, \quad \frac{dP}{dt} = kP.$$

- Second order DEs:

$$\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + x = \sin t.$$

5.6 Introduction to Differential Equations

Introduction and Terminology

- ◇ A **solution** of a given differential equation is a function that satisfies the equation.
- ◇ A **general solution** is a characterization of all possible solutions of the equation.
- ◇ A differential equation is **solved** when we find a general solution.

Consider the differential equation

$$\frac{dy}{dx} = 2x. \quad (11)$$

- $y = x^2$ is a solution to (11).
- A general solution to (11): $y = \int 2x dx = x^2 + C$.

5.6 Introduction to Differential Equations

Introduction and Terminology

Example 28

If $4x - 3y^2 = 10$ and $y \neq 0$, verify that $\frac{dy}{dx} = \frac{2}{3y}$.

5.6 Introduction to Differential Equations

Introduction and Terminology

Example 28

If $4x - 3y^2 = 10$ and $y \neq 0$, verify that $\frac{dy}{dx} = \frac{2}{3y}$.

Differentiate the equation w.r.t x , we get

$$\frac{d}{dx} (4x - 3y^2) = \frac{d}{dx} (10)$$

$$4 - 6y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2}{3y}.$$

5.6 Introduction to Differential Equations

Separable Differential Equations

Definition 29

A differential equation is of the form

$$\frac{dy}{dx} = \frac{f(x)}{g(y)} \quad (12)$$

is so-called **separable differential equation**.

5.6 Introduction to Differential Equations

Separable Differential Equations

Definition 29

A differential equation is of the form

$$\frac{dy}{dx} = \frac{f(x)}{g(y)} \quad (12)$$

is so-called **separable differential equation**.

To solve the equation

- Step 1: separate the variables

$$g(y)dy = f(x)dx. \quad (13)$$

- Step 2: integrate both sides of the equation

$$\int g(y)dy = \int f(x)dx. \quad (14)$$

Example 30

Solve the equation

$$\text{a) } \frac{dy}{dx} = \frac{x}{3y}$$

$$\text{b) } \frac{dy}{dx} = \frac{x}{y} \sqrt{1 - x^2}$$

Example 31

Solve the equation

$$\text{a) } \frac{dy}{dx} = \frac{\cos x}{y(1 - \sin x)}$$

$$\text{b) } \frac{dy}{dx} = \frac{6x^2}{2y + \cos y}$$

Example 32

- (a) Solve the differential equation $\frac{dy}{dx} = \frac{x^2}{y^2}$.
- (b) Find the solution of this equation that satisfies the initial condition $y(0) = 2$.

Applications of Differential Equations

Modeling exponential growth and decay

Exponential change

$$\frac{Q'(t)}{Q(t)} = k \quad \text{or} \quad \frac{dQ}{dt} = kQ(t) \quad (15)$$

$k > 0$: the exponential change is **growth**; $k < 0$: the exponential change is **decay**.

Applications of Differential Equations

Modeling exponential growth and decay

Exponential change

$$\frac{Q'(t)}{Q(t)} = k \quad \text{or} \quad \frac{dQ}{dt} = kQ(t) \quad (15)$$

$k > 0$: the exponential change is **growth**; $k < 0$: the exponential change is **decay**.

The growth/decay equation of a substance is

$$Q(t) = Q_0 e^{kt} \quad (16)$$

where

- $Q(t)$ is the amount of substance present at time t .
- Q_0 is the initial amount of substance.
- k is a constant: $k > 0$: **growth equation**, $k < 0$: **decay equation**.

Example 33

A particular radioactive substance has a half-life of 600 yr. Find k for this substance and determine how much of a 50-g sample will remain after 125 yr.

Applications of Differential Equations

Modeling exponential growth and decay

Carbon dating problem

The ratio of ^{14}C to ^{12}C in a specimen t years after it was alive is approximately

$$R(t) = R_0 e^{kt}, \quad \text{and} \quad k = \frac{\ln \frac{1}{2}}{5,730} \quad (17)$$

where R_0 is the ratio of ^{14}C to ^{12}C in the atmosphere.

Example 34

An archaeologist has found a specimen in which the ratio of ^{14}C to ^{12}C is 20% the ratio found in the atmosphere. Approximately how old is the specimen?

Applications of Differential Equations

Modeling fluid flow through an orifice

Consider a tank that is filled with a fluid being slowly drained through a small, sharp-edged hole in its base. Find the formula for the volume of fluid in the tank at time t ?

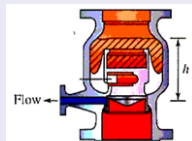


Figure 5.29 The flow of a fluid through a hole of area A_0

Torricelli's law: The rate of flow of fluid

$$\frac{dV}{dt} = -4.8A_0\sqrt{h}, \quad (18)$$

where

- $V = V(t)$ is the **volume of fluid** at time t
- $h = h(t)$ is the **height above the hole** at time t
- A_0 is the **area of the drain hole**

Example 35

A cylindrical tank (with a circular base) is filled with a liquid that is draining through a small circular hole in its base. If the tank is 9 ft high with radius 4 ft, and the drain hole has radius 1 in., how long does it take for the tank to empty?

Applications of Differential Equations

Modeling the motion of a projectile: Escape velocity

Escape velocity

Consider a projectile that is launched with initial velocity v_0 from the planet's surface along a direct line through the center of the planet.

- Find the formula for the velocity v .
- Find the minimal velocity v_0 so that the projectile will escape the planet's gravitational attraction.

The Newton's law of gravitation

If the only force acting on the projectile is due to gravity, then the acceleration of the projectile

$$a = -\frac{gR^2}{s^2} \quad (19)$$

where

- R is the radius of the planet
- g is the acceleration due to gravity at the planet's surface
- s is the distance from the center of the planet.

Find the formula for the velocity v ?

- Step 1: By using the chain rule

$$a = \frac{dv}{dt} = \frac{dv}{ds} \cdot \frac{ds}{dt} = \frac{dv}{ds} v \quad (20)$$

we get the DE

$$\frac{dv}{ds} v = -\frac{gR^2}{s^2}. \quad (21)$$

Find the formula for the velocity v ?

- Step 1: By using the chain rule

$$a = \frac{dv}{dt} = \frac{dv}{ds} \cdot \frac{ds}{dt} = \frac{dv}{ds} v \quad (20)$$

we get the DE

$$\frac{dv}{ds} v = -\frac{gR^2}{s^2}. \quad (21)$$

- Step 2: Solving the equation

$$v^2 = 2gR^2 s^{-1} + C. \quad (22)$$

Find the constant C . Since $v(R) = v_0$ we have

$$C = v_0^2 - 2gR. \quad (23)$$

Thus the formula for the velocity

$$v^2 = 2gR^2s^{-1} + v_0^2 - 2gR. \quad (24)$$

Find the minimal velocity v_0 ?

The projectile will keep moving away from the planet until it reaches a point where $v = 0$.

Thus the formula for the velocity

$$v^2 = 2gR^2s^{-1} + v_0^2 - 2gR. \quad (24)$$

Find the minimal velocity v_0 ?

The projectile will keep moving away from the planet until it reaches a point where $v = 0$.

Since $2gR^2s^{-1} > 0$ for all $s > 0$, we consider two following cases:

- if $v_0^2 - 2gR < 0$ then

$$v = 0 \quad \text{when} \quad s = -\frac{2gR^2}{v_0^2 - 2gR}$$

Thus the formula for the velocity

$$v^2 = 2gR^2s^{-1} + v_0^2 - 2gR. \quad (24)$$

Find the minimal velocity v_0 ?

The projectile will keep moving away from the planet until it reaches a point where $v = 0$.

Since $2gR^2s^{-1} > 0$ for all $s > 0$, we consider two following cases:

- if $v_0^2 - 2gR < 0$ then

$$v = 0 \quad \text{when} \quad s = -\frac{2gR^2}{v_0^2 - 2gR}$$

- if $v_0^2 - 2gR \geq 0$, that is, $v_0 \geq \sqrt{2gR}$ then $v > 0$ for all $s > 0$. Thus the minimal value of v_0 is

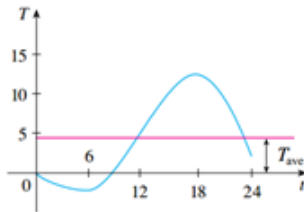
$$v_0 = \sqrt{2gR}.$$

5.7 The mean value theorem for integrals: Average value

The average value of finitely many numbers $y_1, y_2, \dots, y_n$:

$$y_{ave} = \frac{y_1 + y_2 + \dots + y_n}{n}.$$

How do we compute the average temperature during a day if infinitely many temperature readings are possible?



AVERAGE VALUE

If f is continuous on $[a, b]$, the **average value** of f on $[a, b]$, denote by f_{ave} , is given by

$$f_{ave} = \frac{1}{b-a} \int_a^b f(x) dx. \quad (25)$$

Example 36

Suppose a study suggests that between the hours of 1:00 PM and 4:00 PM on a normal weekday the speed of the traffic at a certain expressway exit is modeled by the formula

$$S(t) = 2t^3 - 21t^2 + 60t + 20$$

kilometers per hour, where t is the number of hours past noon. Compute the average speed of the traffic between hours of 1:00 PM and 4:00 PM.

Proof of the Formula (25)

- Step 1: Divide the interval $[a, b]$ into equal subintervals, each with length $\Delta x = \frac{b-a}{n}$.
- Step 2: Choose points $x_1^*, x_2^*, \dots, x_n^*$, in successive subintervals and approximate the average value of f , denote by f_{ave}

$$f_{ave} \approx \frac{f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)}{n} \quad (26)$$

Simple calculation yields

$$f_{ave} \approx \frac{1}{b-a} [f(x_1^*)\Delta x + f(x_2^*)\Delta x + \dots + f(x_n^*)\Delta x]$$

- Step 3: Let $n \rightarrow \infty$, we get

$$f_{ave} = \frac{1}{b-a} \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*)\Delta x = \frac{1}{b-a} \int_a^b f(x)dx.$$

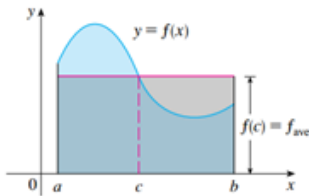
Theorem 37 (The Mean Value Theorem for Integrals)

If f is continuous on $[a, b]$, there exists a number $c \in [a, b]$ such that

$$f(c) = f_{ave} = \frac{1}{b-a} \int_a^b f(x) dx. \quad (27)$$

The **geometric interpretation** of the Mean Value Theorem for Integrals

$$\int_a^b f(x) dx = f(c)(b-a)$$



Example 38

Suppose that during a typical winter day in Minneapolis, the temperature (in degree Celsius) x hours after midnight is modeled by the formula

$$T(x) = 2 - \frac{1}{7} (x - 13)^2.$$

Find the average temperature over the time period from 2:00 AM to 2:00 PM, and find a time when the average temperature actually occurs.

5.8 Numerical integration: The trapezoidal rule and Simpson's rule

Approximate by Rectangle

Let f be continuous on $[a, b]$. The **rectangle rule** is

$$\int_a^b f(x) dx \approx [f(x_0) + f(x_1) + \cdots + f(x_{n-1}) + f(x_n)] \Delta x \quad (28)$$

where $\Delta x = \frac{b-a}{n}$, $x_k = a + k\Delta x$.

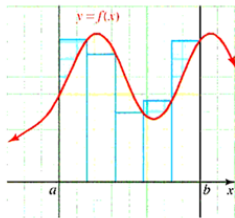


Figure 5.36 Interactive Approximation by rectangles

5.8 Numerical integration: The trapezoidal rule and Simpson's rule

Trapezoidal Rule

Let f be continuous on $[a, b]$. The **trapezoidal rule** is

$$\int_a^b f(x) dx \approx \frac{1}{2} [f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)] \Delta x \quad (29)$$

where $\Delta x = \frac{b-a}{n}$, $x_k = a + k\Delta x$.

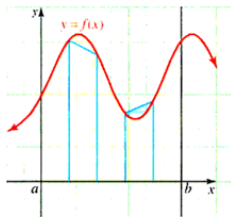


Figure 5.37 Interactive Approximation by trapezoids

Sketch the Idea

- Step 1: Divide the interval $[a, b]$ into equal subintervals, each with length

$$\Delta x = \frac{b - a}{n}$$

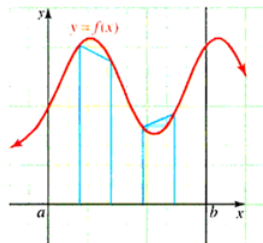


Figure 5.37 Interactive Approximation by trapezoids

- Step 2: Let T_n be the sum of the areas of n trapezoids, we see that

$$\begin{aligned} T_n &= \frac{1}{2} [f(x_0) + f(x_1)] \Delta x + \frac{1}{2} [f(x_1) + f(x_2)] \Delta x \\ &\quad + \cdots + \frac{1}{2} [f(x_{n-1}) + f(x_n)] \Delta x \\ &= \frac{1}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)] \Delta x \end{aligned}$$

5.8 Numerical integration: The trapezoidal rule and Simpson's rule

Simpson's Rule

Let f be continuous on $[a, b]$. The **Simpson's rule** is

$$\int_a^b f(x) dx \approx \frac{1}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)] \Delta x \quad (30)$$

where $\Delta x = \frac{b-a}{n}$, $x_k = a + k\Delta x$, k an integer and n an even integer.

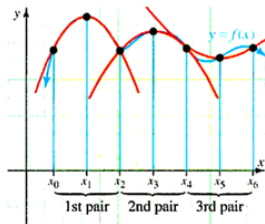


Figure 5.40 Approximation using parabolas